

Appendix A2 - Ningaloo Marine Park (National Park Zone) Natural Values Report

North-west Marine Parks Network

Table of contents

Natural Values	1
Sampling summary	2
Benthic ecosystem extent	3
Observations	3
Dominant habitat types	4
Habitat distribution and probability	5
Functional reef distribution and probability	6
Benchmarks of benthic natural values	7
Sand	7
Sessile invertebrates	8
Benchmarks of demersal fish natural values	9
Survey effort	9
Species richness	10
Total abundance	12
Large Reef Fish Index*	14
Reef fish thermal index	16

Natural Values

Understanding of natural values

Sampling summary

Table A2.1: Summary of monitoring campaigns in the study area, showing the survey dates, sampling method, management zones sampled, and the number of samples with each data type. NPZ = National Park Zone, HPZ = Habitat Protection Zone, MUZ = Multiple Use Zone, SPZ = Special Purpose Zone.

Date	Campaign name	Method	Areas	Samples	Habitat	Fish	
					Count	Count	Length
Aug 2019	2019-08 Ningaloo-deep stereoBRUVs	BRUV	Coastal waters, MUZ, NPZ, RUZ	132	132	123	117
May 2021	2021-05 PtCloates stereo-BRUVs	BRUV	MUZ, NPZ	25	25	25	24
May 2021	2021-05 PtCloates BOSS	BOSS	MUZ, NPZ	39	39	-	-
Aug-Sep 2021	2021-08 Pt-Cloates Flasher-BOSS	BOSS	NPZ	67	67	-	-
Aug-Sep 2021	2021-08 Pt-Cloates stereo-BRUVs	BRUV	NPZ	47	47	45	44
Sep 2021	2021-08 Pt-Cloates Squid-BOSS	BOSS	NPZ	30	30	-	-
May 2022	2022-05 PtCloates Naked-BOSS	BOSS	NPZ	74	74	-	-
May 2022	2022-05 PtCloates Squid-BOSS	BOSS	NPZ	74	65	-	-
May 2022	2022-05 PtCloates stereo-BRUVs	BRUV	NPZ	64	64	62	62
May 2024	DR2024-02 CSIRO Ningaloo stereoBRUVs	BRUV	MUZ, NPZ, RUZ	71	65	63	55

Benthic ecosystem extent

Observations

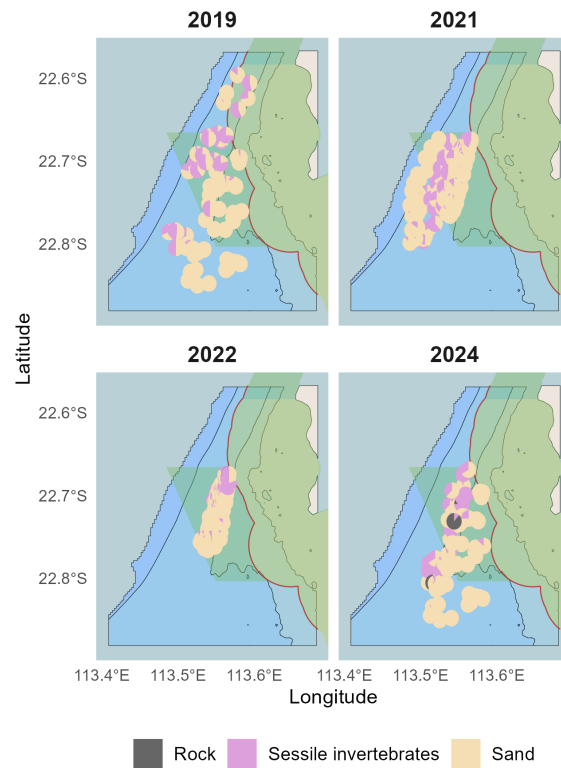


Figure A2.1: Habitat classes observed in spatially balanced stereo-BRUV and drop-camera imagery visualised as spatial pie charts. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Dominant habitat types

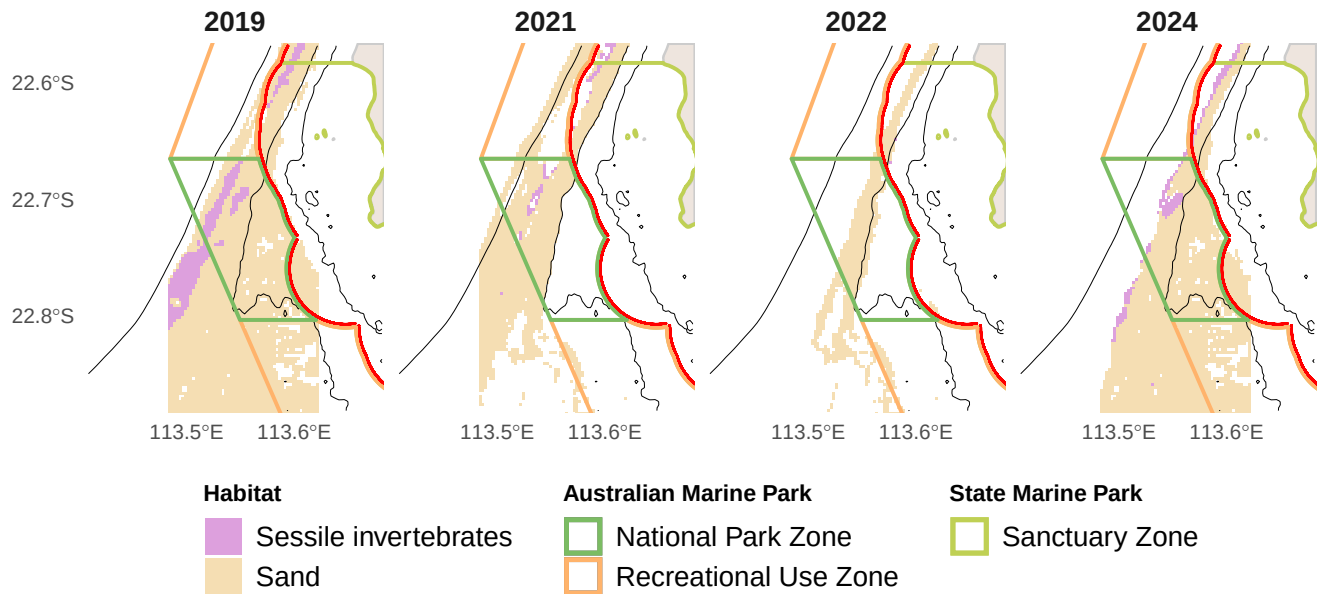


Figure A2.2: Predicted dominant habitat type by year from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Habitat distribution and probability

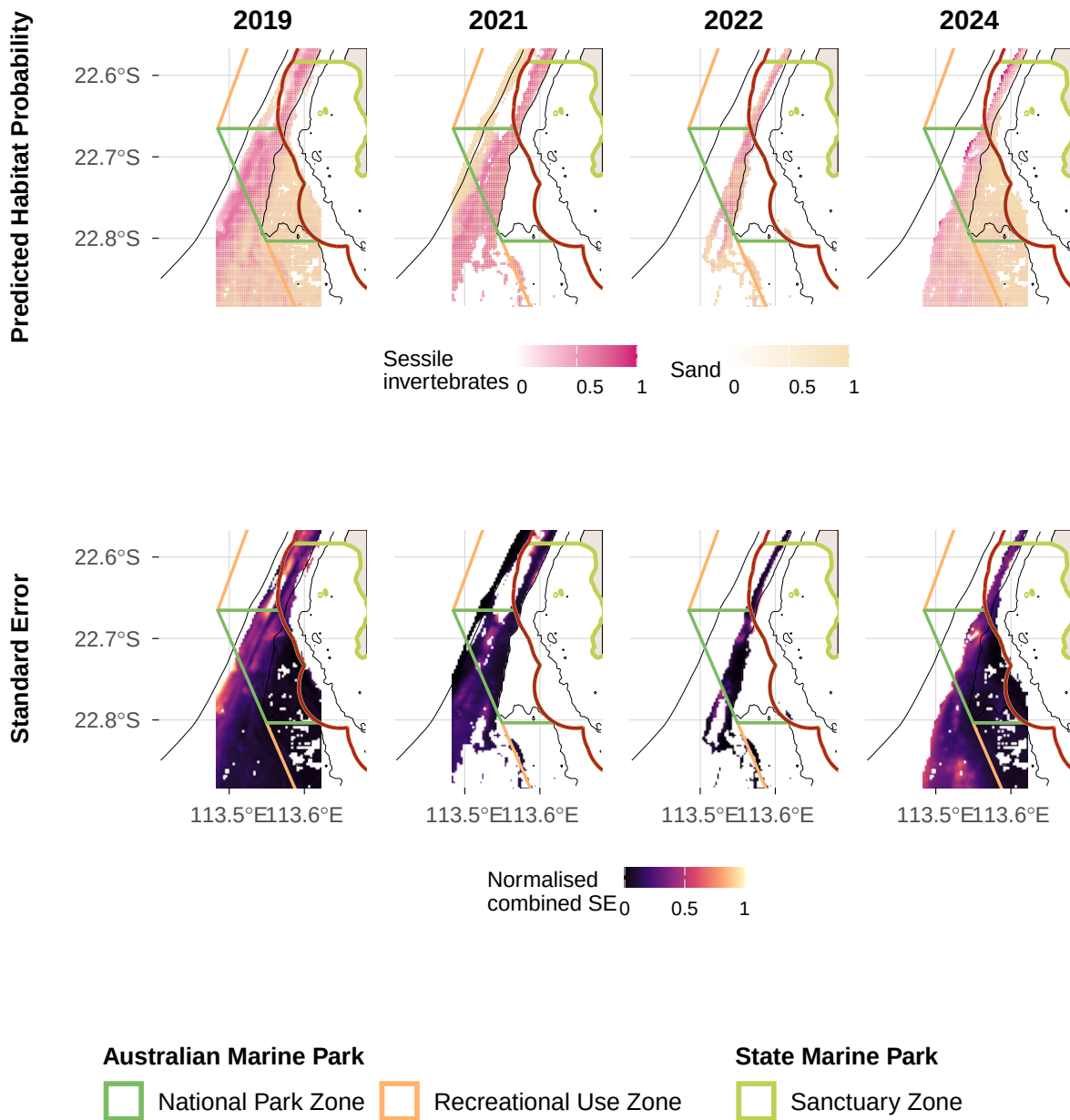


Figure A2.3: Predicted habitat distribution probability and uncertainty by year from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Funcional reef distribution and probability

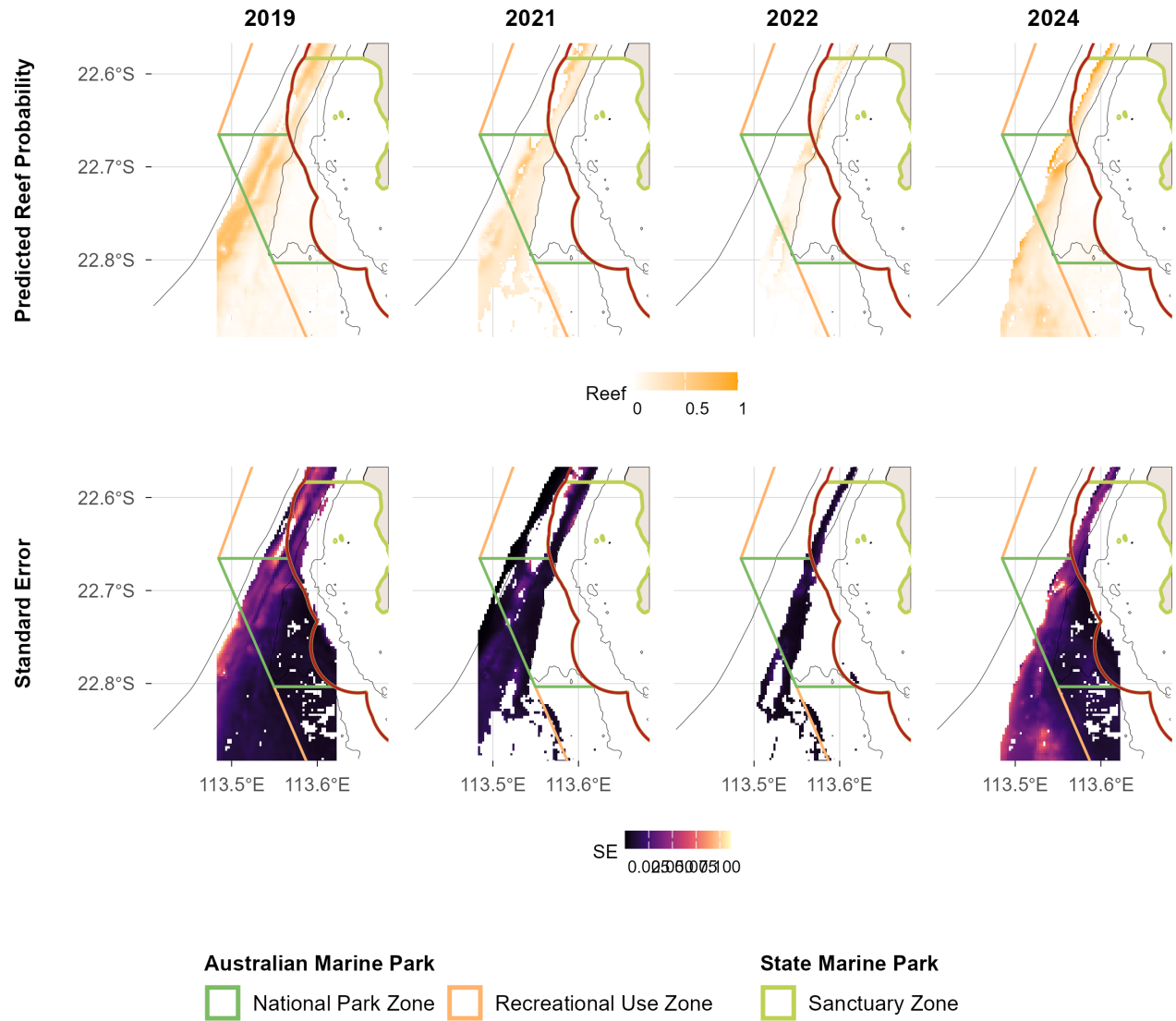


Figure A2.4: Predicted reef distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Benchmarks of benthic natural values

Sand

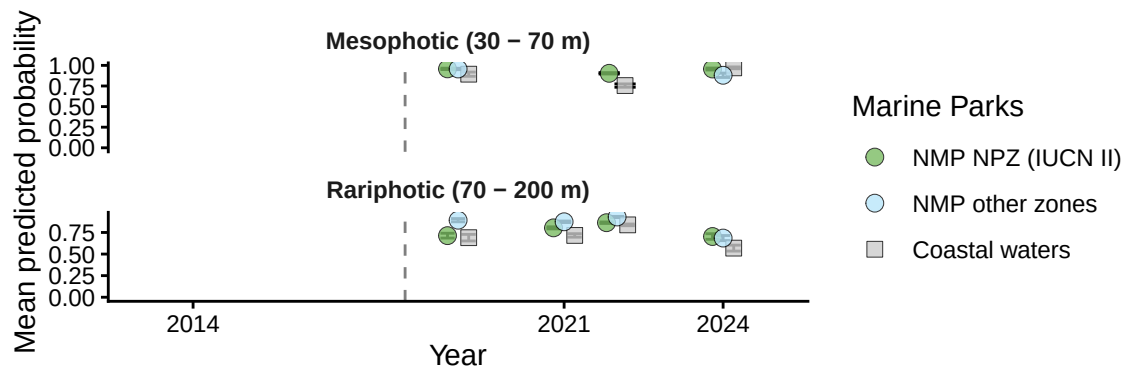


Figure A2.5.1: Sand: benchmark plots by zone and depth contour from stereo-BRUV groundtruthing. NMP = Ningaloo Marine Park (Commonwealth), NMP (State) = Ningaloo Marine Park (State).

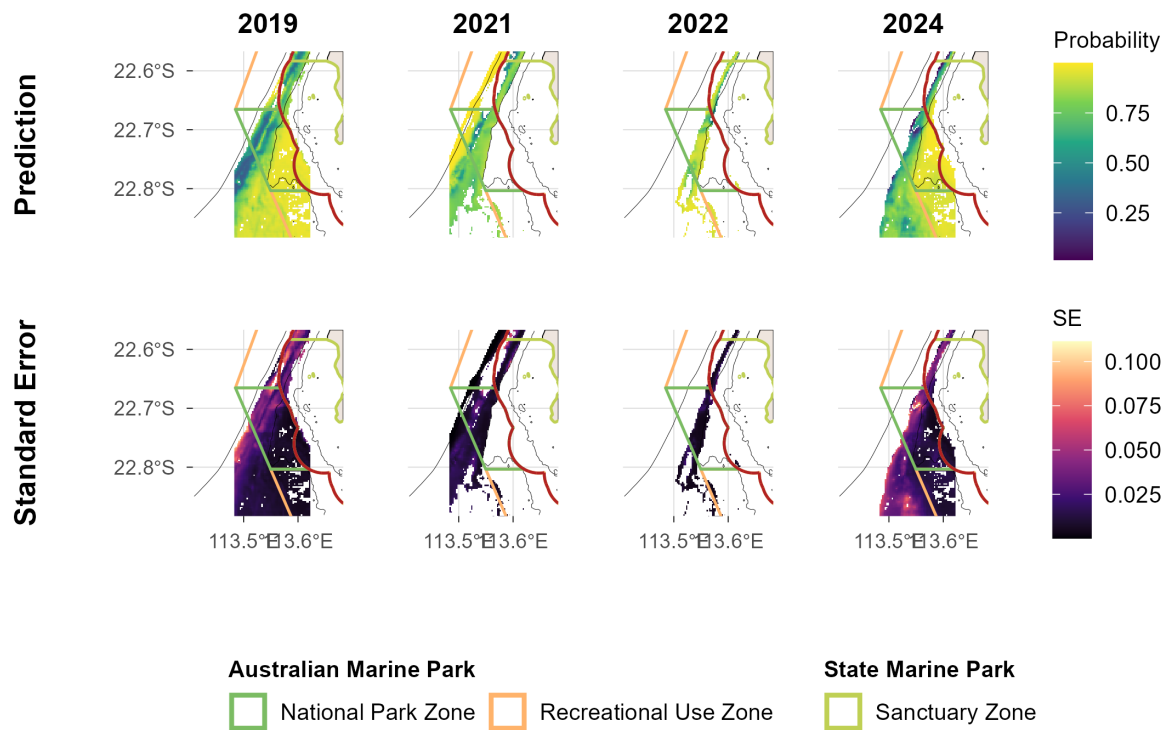


Figure A2.5.2: Sand: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Sessile invertebrates

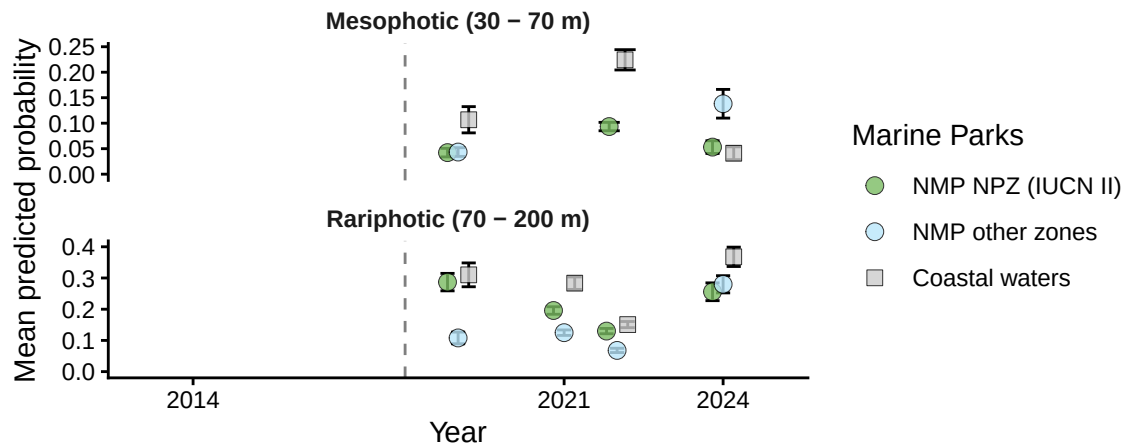


Figure A2.6.1: Sessile invertebrates: benchmark plots by zone and depth contour from stereo-BRUV groundtruthing. NMP = Ningaloo Marine Park (Commonwealth), NMP (State) = Ningaloo Marine Park (State).

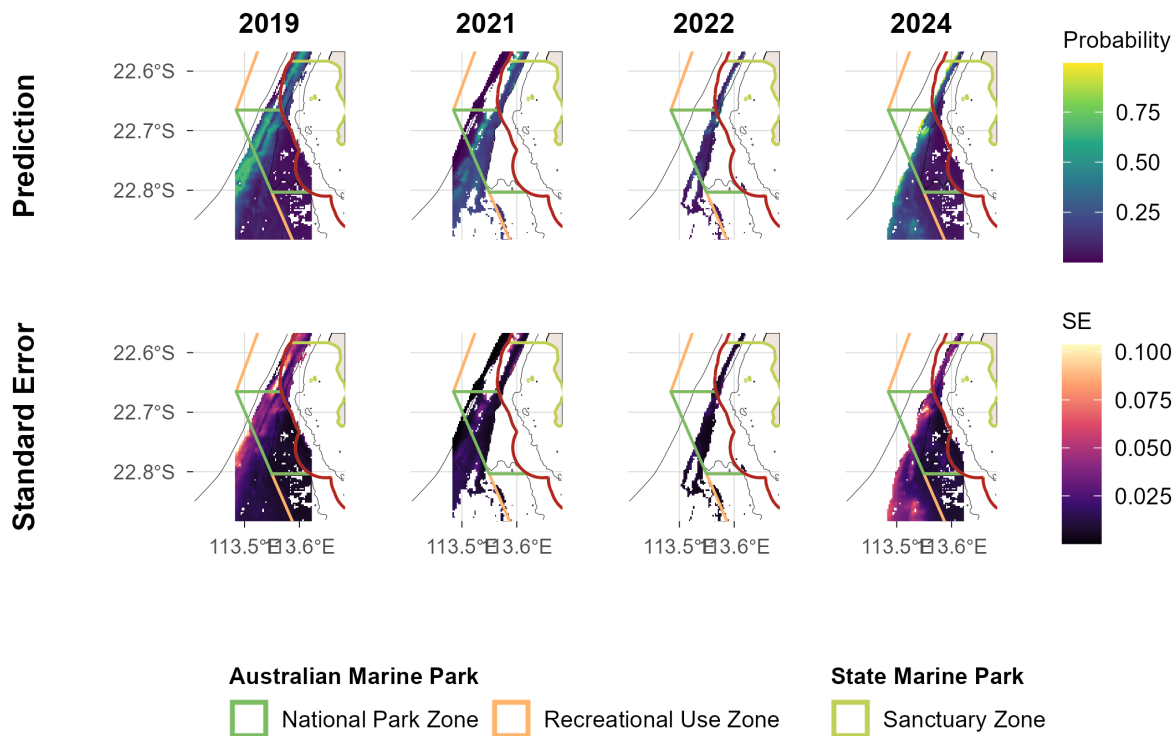
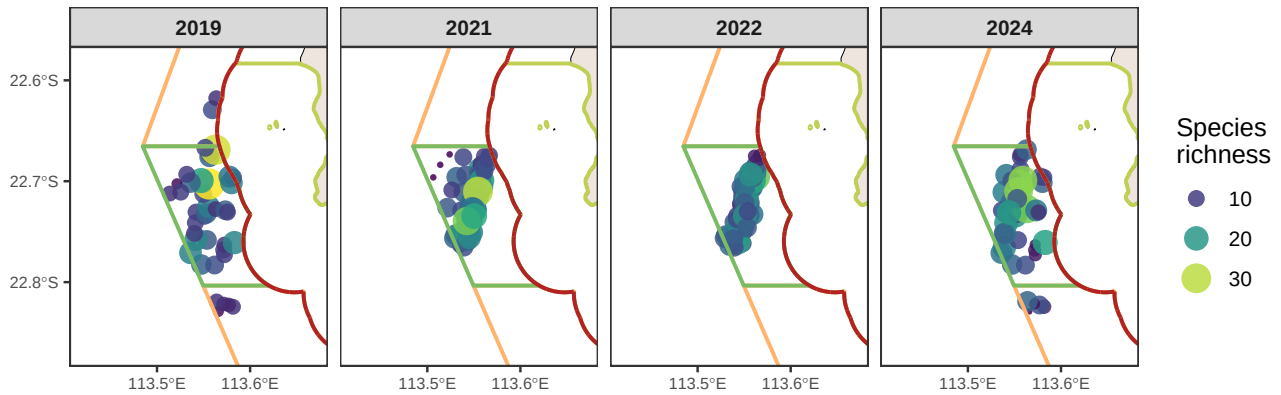


Figure A2.6.2: Sessile invertebrates: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

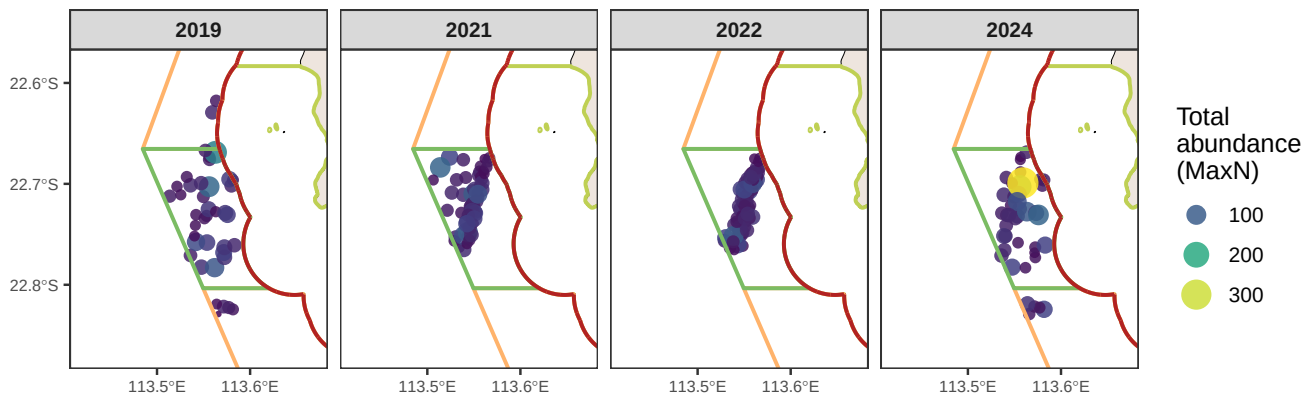
Benchmarks of demersal fish natural values

Survey effort

a)



b)



Australian Marine Park

National Park Zone

Recreational Use Zone

State Marine Park

Sanctuary Zone

Figure A2.7: Spatial distribution of observed species richness and total abundance (MaxN) per stereo-BRUV drop, pooled across all deployments. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters.

Species richness

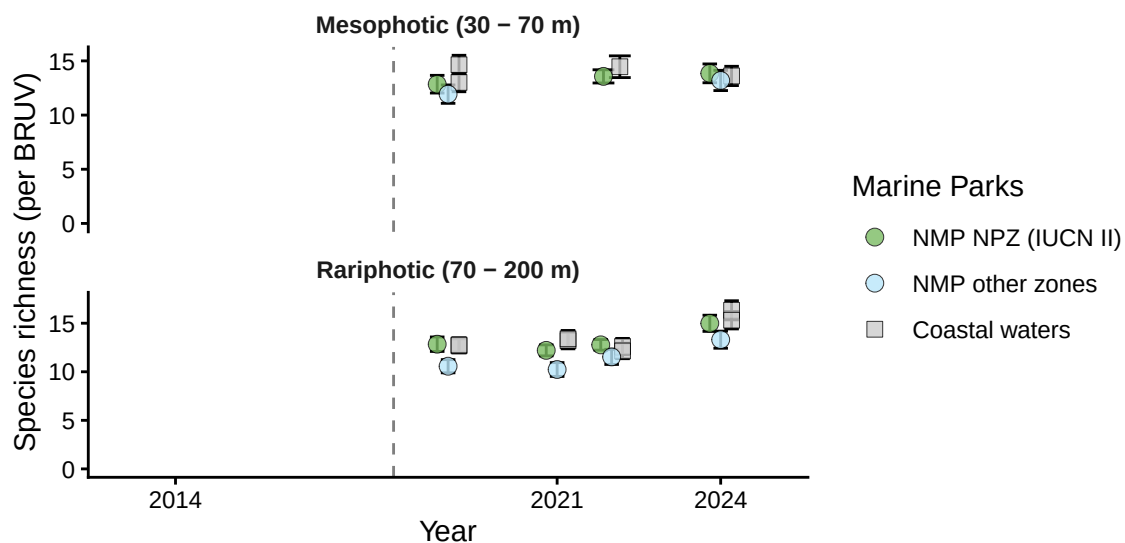


Figure A2.8.1: Demersal fish - Species richness: benchmark plots by zone and depth contour from stereo-BRUV groundtruthing. NMP = Ningaloo Marine Park (Commonwealth).

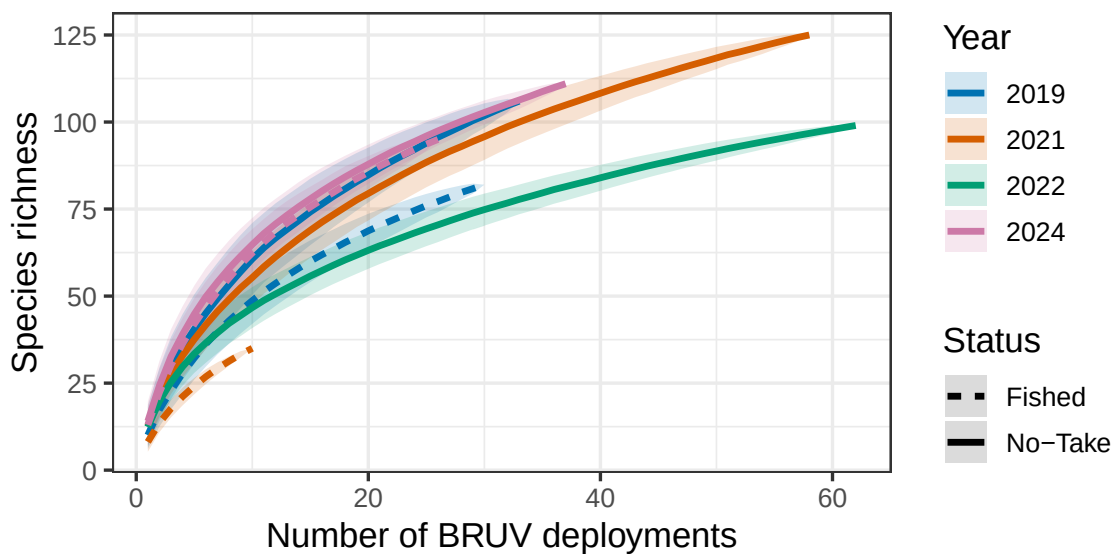


Figure A2.8.2: Demersal fish - Species richness: species accumulation curve by year from stereo-BRUV groundtruthing. NMP = Ningaloo Marine Park (Commonwealth).

Species richness

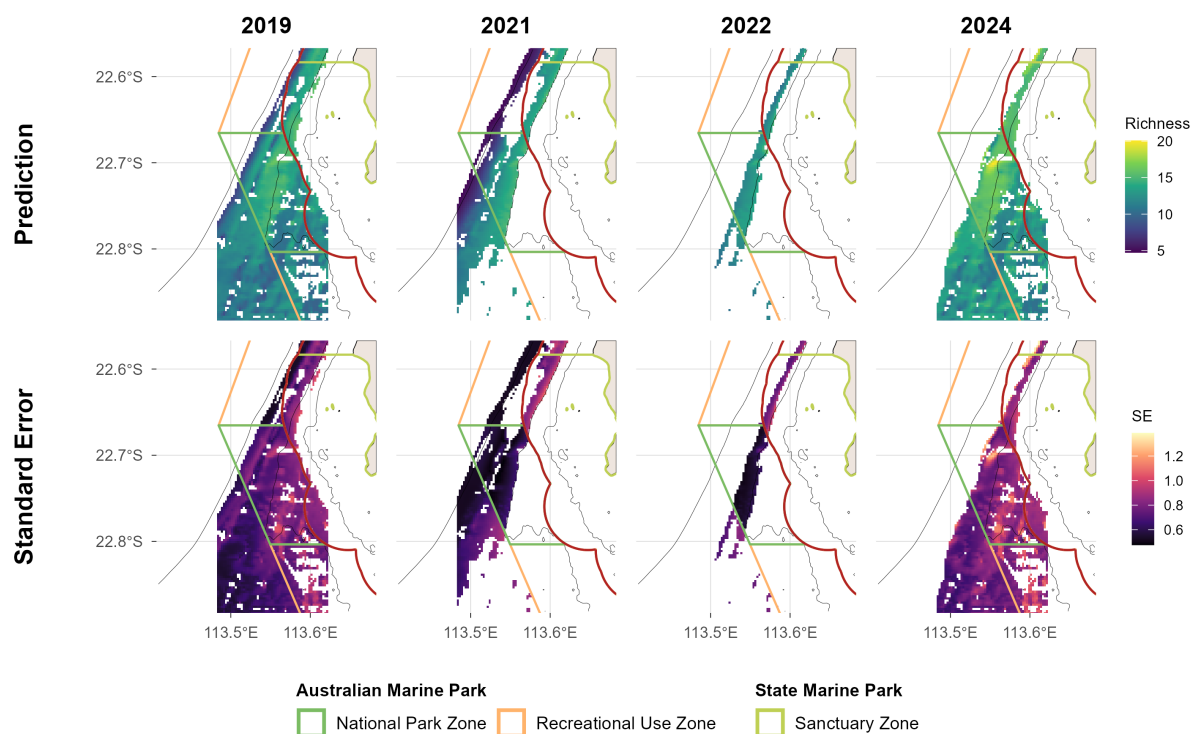


Figure A2.8.3: Demersal fish - Species richness: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Total abundance

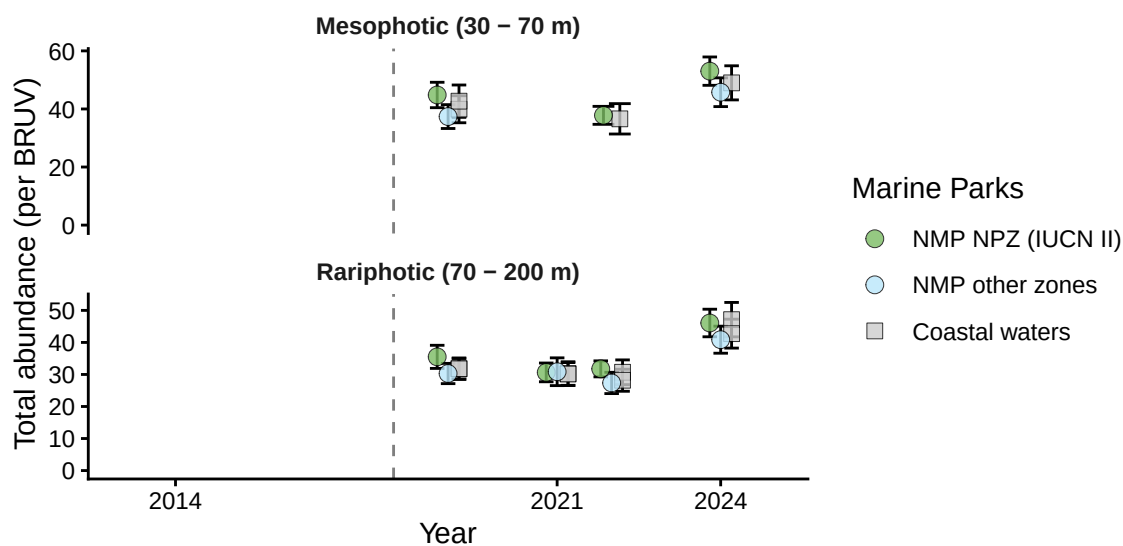


Figure A2.9.1: Demersal fish - Total abundance: benchmark plots by zone and depth contour from stereo-BRUV groundtruthing. NMP = Ningaloo Marine Park (Commonwealth).

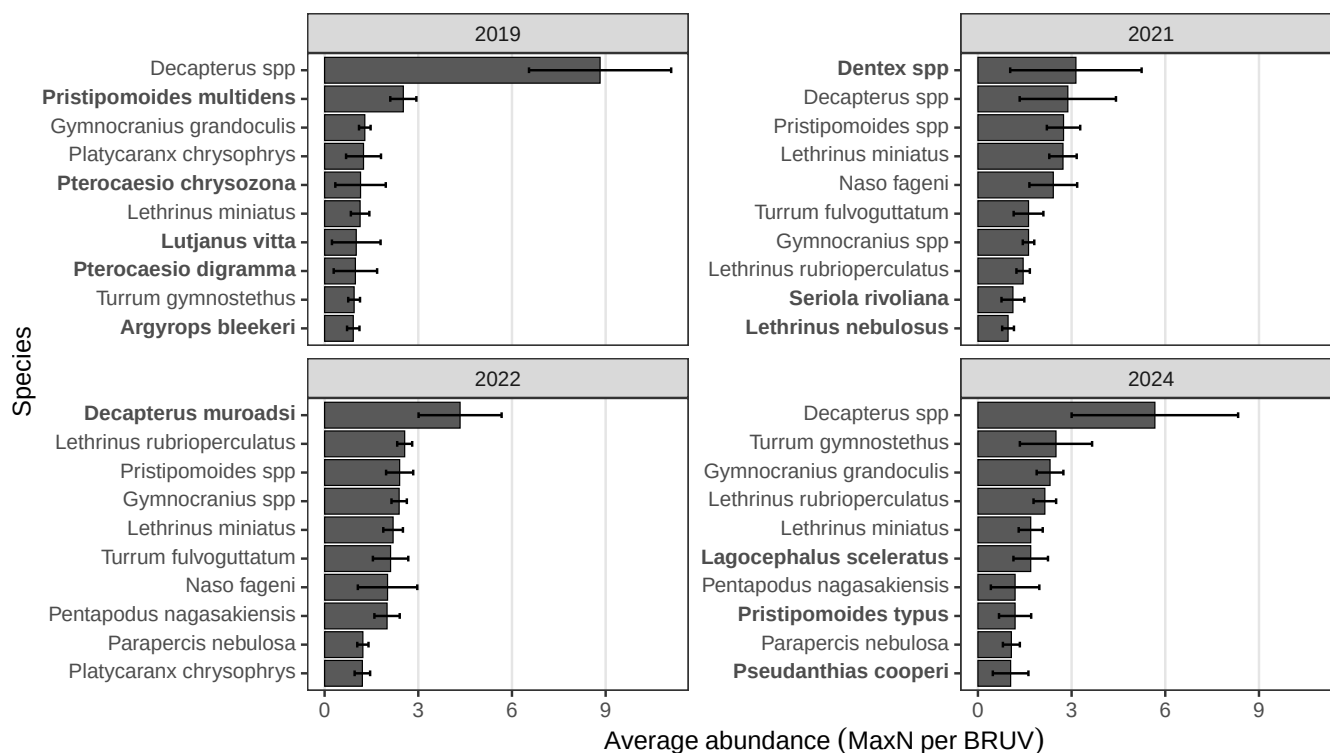


Figure A2.9.2: Demersal fish - Total abundance: top 10 most abundant species (pooled across deployments) from stereo-BRUV groundtruthing. NMP = Ningaloo Marine Park (Commonwealth).

Total abundance

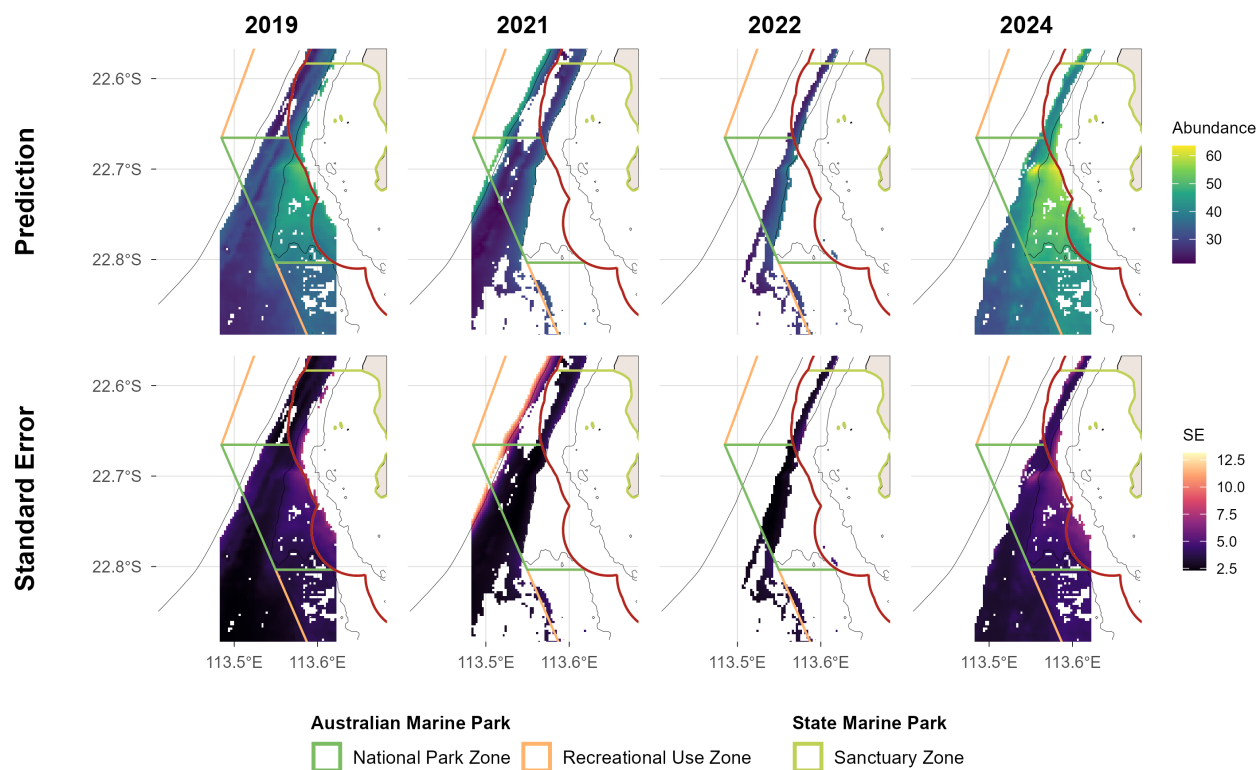


Figure A2.9.3: Demersal fish - Total abundance: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Large Reef Fish Index*

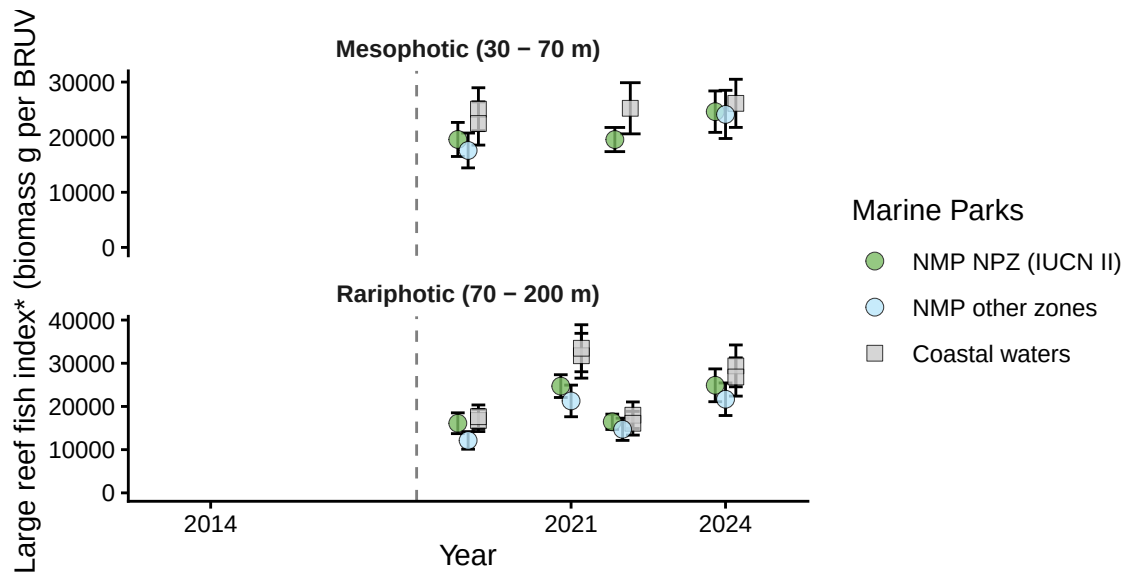


Figure A2.10.1: Demersal fish - Large Reef Fish Index*: benchmark plots by zone and depth contour from stereo-BRUV groundtruthing. NMP = Ningaloo Marine Park (Commonwealth). *Large Reef Fish Index representing the biomass of all fish greater than 20 cm (B20) has been adapted here for stereo-BRUV data by removing sharks and rays.

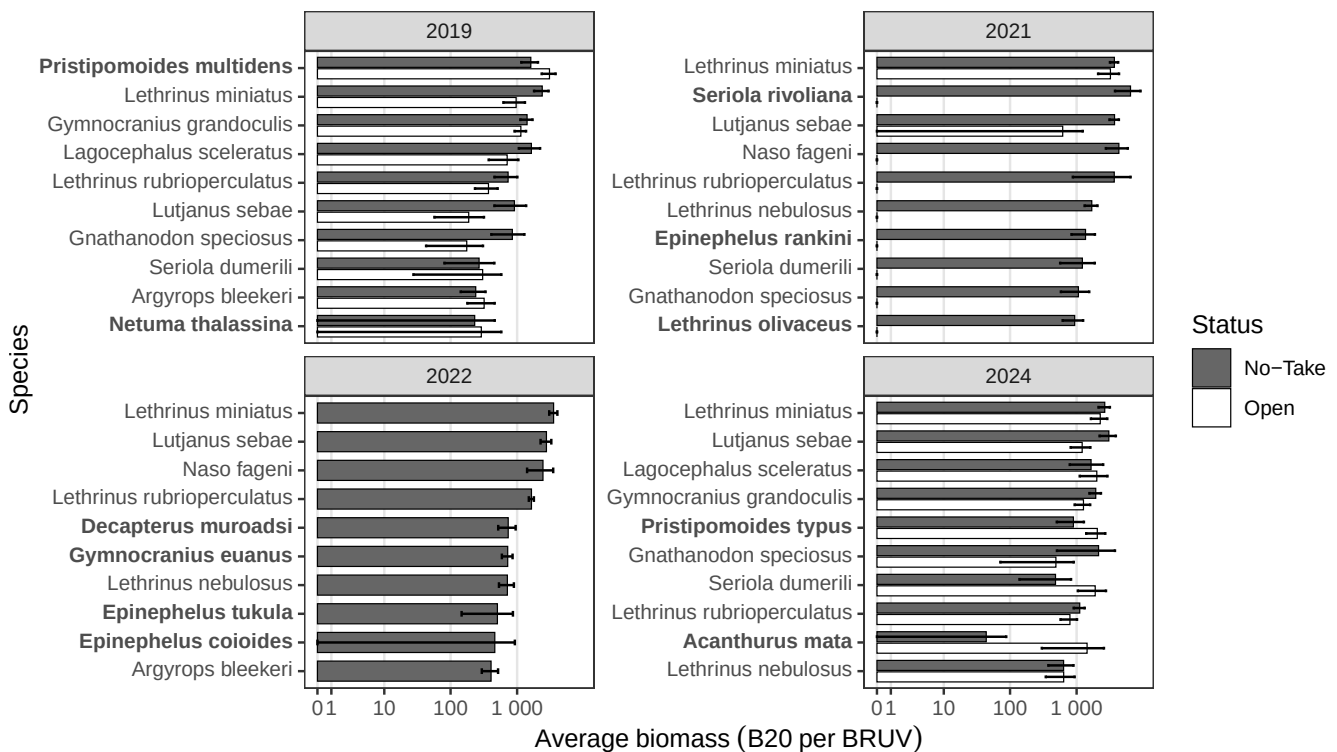


Figure A2.10.2: Demersal fish - Large Reef Fish Index*: top 10 greatest biomass species (pooled across deployments), split by Fished and No-Take status, from stereo-BRUV groundtruthing. NMP = Ningaloo Marine Park (Commonwealth).

Large Reef Fish Index*

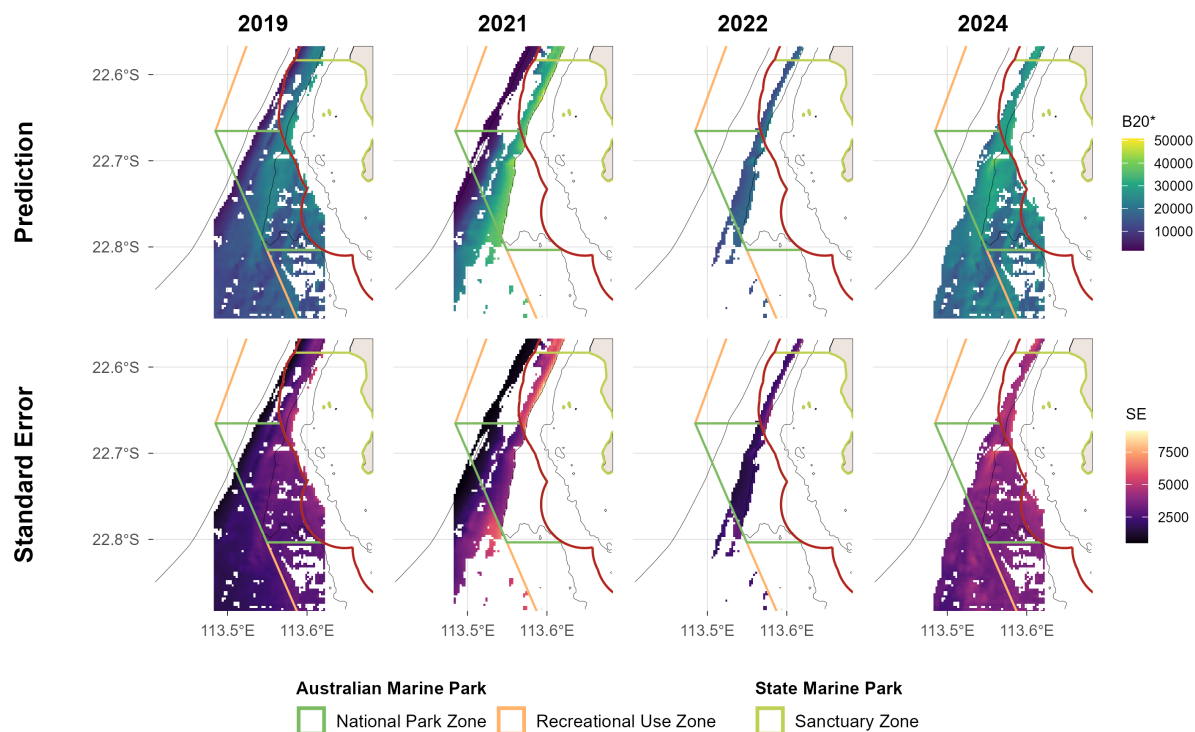


Figure A2.10.3: Demersal fish - Large Reef Fish Index*: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Reef fish thermal index

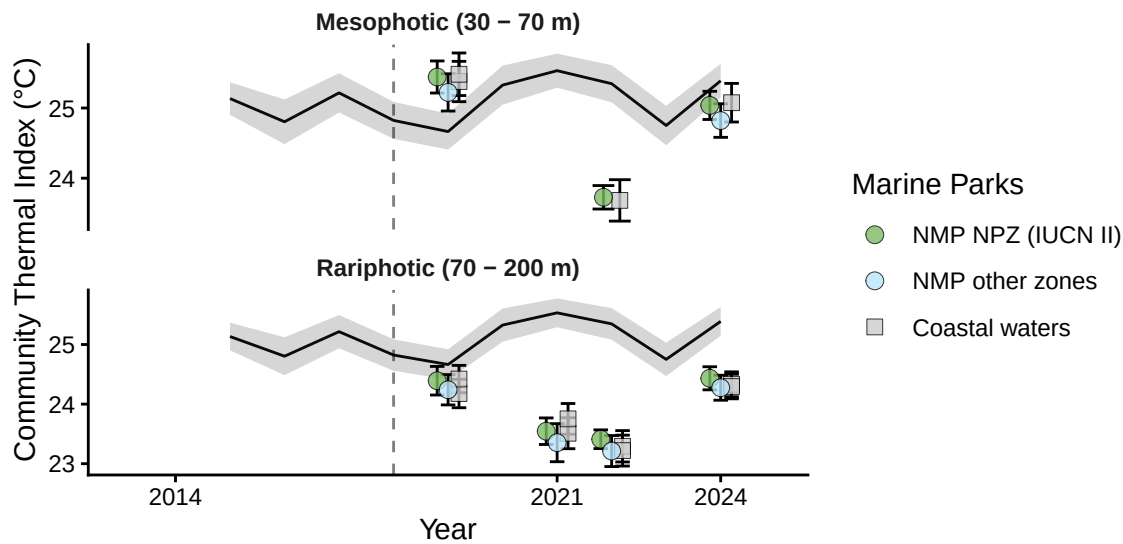


Figure A2.11.1: Demersal fish - Reef Fish Thermal Index: benchmark plots by zone and depth contour, with monthly sea surface temperature (black line, grey band = SD) overlaid, from stereo-BRUV groundtrawling. NMP = Ningaloo Marine Park (Commonwealth).

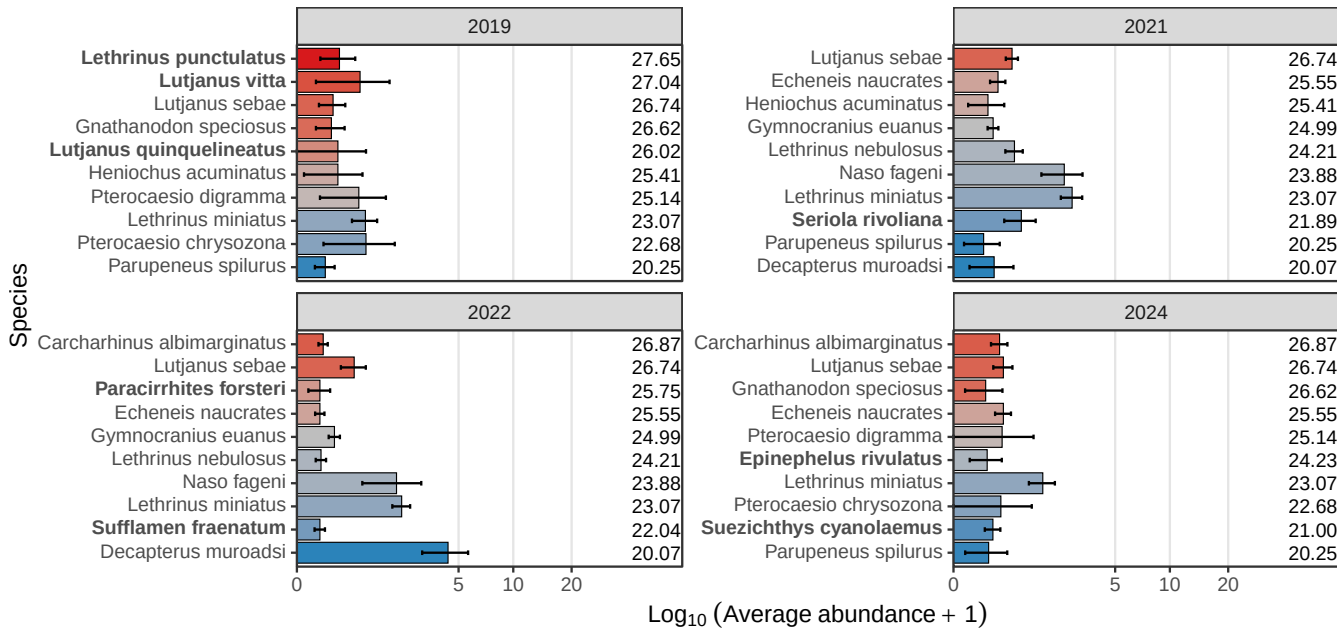


Figure A2.11.2: Demersal fish - Reef Fish Thermal Index: species thermal niche of top 10 most abundant species (pooled across deployments) from stereo-BRUV groundtrawling. NMP = Ningaloo Marine Park (Commonwealth).

Reef fish thermal index

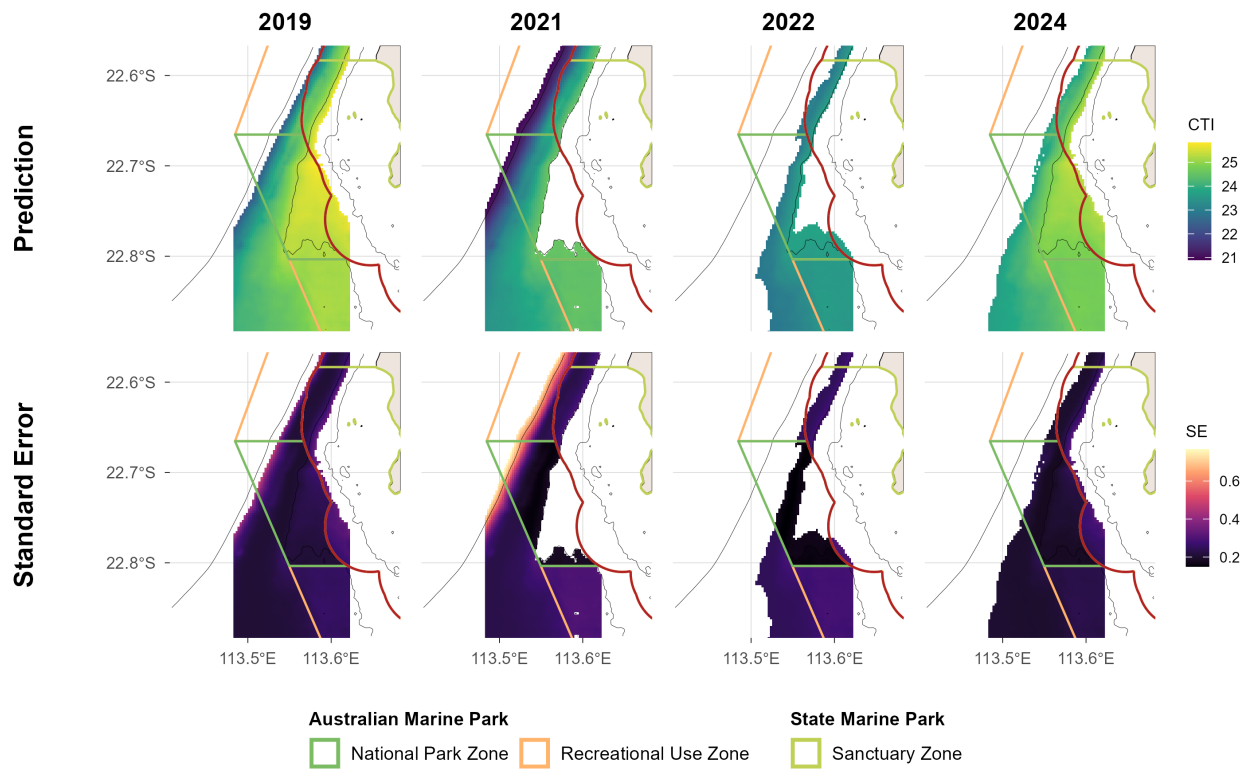


Figure A2.11.3: Demersal fish - Reef Fish Thermal Index: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.